Accuracy–Interpretability Trade-offs in Exoplanet Transit Modelling, and a Label-Free Physics-Informed Estimator of Planetary Radii

BHAVYA GONDRA*
Nexus Labs, Palo Alto, California, USA
AKHIL MUVVALA*
Nexus Labs, Palo Alto, California, USA
MILAN PETROV*
Nexus Labs, Palo Alto, California, USA
HARISH SENTHIL*
Nexus Labs, Palo Alto, California, USA
AYAN SHARMA*
Nexus Labs, Palo Alto, California, USA
MANAV SHARMA*
Nexus Labs, Palo Alto, California, USA
PRANAVAN SIVARAMAN*
Nexus Labs, Palo Alto, California, USA
ADHRIT SOMAYAJULA*
Nexus Labs, Palo Alto, California, USA
SRIKANTH SAMY
University of California, Berkeley, Berkeley, California, USA
ssamy@berkeley.edu
ISHAAN GUPTA
Stanford University, Stanford, California, USA
ishaangupta@stanford.edu

Transit photometry poses two distinct machine-learning problems: deciding whether a light curve contains a planetary transit, and measuring the planet's radius ratio R_p/R_* from it. We compare seven model families on both tasks using simulated light curves with exactly known parameters, and score every model on a second axis besides accuracy—the fidelity with which a purely additive surrogate reproduces its output, an explicit and reproducible operationalisation of one dimension of interpretability. The two axes trade off: on detection, the most accurate model (an MLP, AUC = 0.975) has the second-lowest additive fidelity ($\mathcal{A}$ = 0.573), while the fully readable linear model reaches AUC = 0.946 at $\mathcal{A}$ = 0.999—a 0.029 AUC premium for near-total transparency. We then introduce a physics-informed Kolmogorov–Arnold network that predicts transit parameters and is trained *only* to reconstruct the observed flux through a differentiable Mandel–Agol forward model, never seeing a radius label. It attains MAE = 0.0100 in R_p/R_*, within 24% of the best fully supervised model, and occupies a distinct region of the Pareto plane. Applied zero-shot to twelve confirmed Kepler planets it recovers radii with full-sample MAE = 0.023. We report that full-sample figure as the primary real-data result and show that the label-free reconstruction-residual screen used to identify model mismatch improves it only to 0.018—and rejects one fold whose radius was in fact recovered to within 0.001, demonstrating that reconstruction quality is not a reliable proxy for parameter accuracy.

CCS Concepts: • **Applied computing: Astronomy** • **Computing methodologies: Machine learning** • **Computing methodologies: Neural networks**

*These authors contributed equally to this work.

Additional Key Words and Phrases: Exoplanets, interpretable machine learning, Kolmogorov–Arnold networks, physics-informed learning, transit photometry, model comparison.

1 INTRODUCTION

When a planet crosses the disc of its host star along our line of sight, it occults a fraction of the stellar flux and the star appears momentarily fainter. The resulting periodic dip in a light curve—a time series of stellar brightness—is the most productive exoplanet-detection signal known, and it is the mechanism behind the bulk of the confirmed planet catalogue [1, 13].

Machine learning entered this field primarily as a detector. Deep convolutional classifiers now rival or exceed hand-tuned pipelines at separating genuine transits from instrumental and astrophysical false positives [10, 15]. But detection is only half of the scientific product. The dip's depth encodes the planet's size, its duration encodes the orbital geometry, and it is those physical parameters, not the detection flag, that enter a population study. A model that classifies well but whose reasoning cannot be inspected is a weaker scientific instrument than one whose inferred quantities can be checked against physics [14].

This paper studies both halves, and studies them along two axes.

1.1 Contributions

(1) A controlled comparison of seven model families on transit detection and on radius-ratio regression, using a simulator whose ground-truth parameters are known exactly, with a common training and scoring interface for every model.

(2) An explicit, reproducible interpretability score—additive-surrogate fidelity $\mathcal{A}$—applied identically to all models, and a Pareto analysis that makes the accuracy–transparency exchange rate quantitative rather than rhetorical.

(3) A physics-informed KAN that infers R_p/R_*, impact parameter, scaled semi-major axis, and epoch by reconstructing the light curve through a differentiable Mandel–Agol model, using *no* parameter labels at any point in training.

(4) A zero-shot transfer test on twelve confirmed Kepler planets in which the full sample is the primary reported result, and an explicit demonstration that the residual-based quality screen used to flag model mismatch is not aligned with parameter accuracy.

2 FORWARD MODEL AND SIMULATED DATA

2.1 Transit physics

For a planet of radius ratio $\rho = R_p/R_*$ on a circular orbit of scaled semi-major axis a/R_* and impact parameter b, the projected star–planet separation at orbital phase ϕ is

$$d(\phi) = \sqrt{\left(\frac{a}{R_*}\right)^2 \sin^2(2\pi\phi) + b^2\cos^2(2\pi\phi)}. \tag{1}$$

The stellar disc is limb darkened: it is brighter at the centre than at the edge. We use the standard quadratic law [2, 5]

$$\frac{I(\mu)}{I(1)} = 1 - u_1(1-\mu) - u_2(1-\mu)^2, \qquad \mu = \sqrt{1-r^2}, \tag{2}$$

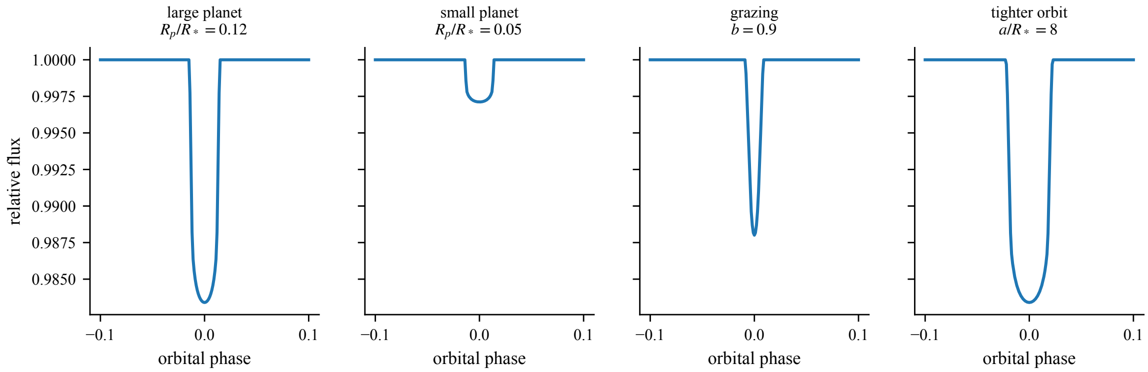

Fig. 1. Transit profiles from Eq. (3). Depth scales approximately as ρ^2; a grazing geometry ($b \to 1$) rounds the profile and destroys the flat bottom; a tighter orbit shortens the crossing. Depth and duration are the two observables from which ρ must be inferred.

with $(u_1, u_2) = (0.3, 0.2)$ throughout. Decomposing the disc into N_r concentric annuli of intensity I_k and computing the occulted area $O(d, R_k, \rho)$ of each analytically [9], the normalised flux is

$$F(\phi) = 1 - \frac{\sum_{k=1}^{N_r} I_k \left[O(d(\phi), R_k, \rho) - O(d(\phi), R_{k-1}, \rho)\right]}{\sum_{k=1}^{N_r} I_k\, \pi\left(R_k^2 - R_{k-1}^2\right)}. \tag{3}$$

Equation (3) is differentiable in $(\rho, b, a/R_*, t_0)$, which is what makes the physics-informed estimator of Section 5 trainable. The reference implementation is validated against `batman` [6]. Figure 1 shows how each parameter reshapes the dip.

2.2 Labelled simulation

Each example is a $L = 201$-point segment spanning orbital phase ± 0.10. Positives draw $\rho \sim \mathcal{U}(0.03, 0.15)$, $b \sim \mathcal{U}(0, 0.95)$, $a/R_* \sim \mathcal{U}(8, 18)$, and a small epoch jitter, then add Gaussian noise of per-point scale $\sigma = 10^u$, $u \sim \mathcal{U}(-3.3, -2.0)$. Negatives are noise-only; half additionally carry a low-frequency sinusoid imitating intrinsic stellar variability, which forces the detector to learn the transit *shape* rather than merely “something changed”. The transit signal-to-noise ratio is

$$\mathrm{SNR} = \frac{\delta}{\sigma/\sqrt{n_{\mathrm{in}}}}, \qquad \delta = 1 - \min_{\phi} F(\phi), \tag{4}$$

with n_{in} the number of in-transit samples implied by the analytic duration. The training set holds 8,000 segments (4,006 positive), the disjoint test set 3,000 (1,483 positive).

Preprocessing differs by task, and the difference is substantive. For detection each curve is standardised,

$$\tilde{x}_i = \frac{x_i - \mathrm{median}(x)}{\mathrm{std}(x)}, \tag{5}$$

which discards absolute depth—appropriate, because depth is a nuisance for the presence/absence question. For characterisation, depth *is* the signal, so we subtract the median without rescaling.

3 MODELS AND THE INTERPRETABILITY AXIS

3.1 The seven families

All models consume the same 201-point vector and expose a common `fit`/`score` interface: regularised linear (logistic for detection, ridge for regression), random forest, gradient-boosted trees, a three-layer MLP, a 1-D CNN with two convolution–pooling blocks, a bidirectional LSTM, and a Kolmogorov–Arnold network (KAN) [8].

The KAN merits a note because it is the reason interpretability is measurable rather than assumed. A conventional network applies fixed nonlinearities to learned linear combinations; a KAN instead learns a spline ψ_j per input coordinate and sums them,

$$g(\mathbf{x}) = \sum_{j=1}^{L} \psi_j(x_j), \tag{6}$$

which is a generalised additive model [4] with learned shape functions. After training, ψ_j can simply be plotted: the model's dependence on each phase bin is read off directly, with no post-hoc approximation [12].

3.2 Additive-surrogate fidelity

Interpretability has no canonical scalar. We therefore measure one specific, well-defined property: how closely the model behaves like a sum of per-coordinate functions over an independent probe distribution. Given a model output $s(\mathbf{x})$—mapped to the logit scale for classifiers—we fit a cubic-spline additive surrogate

$$\hat{s}(\mathbf{x}) = \beta_0 + \sum_{j=1}^{L} h_j(x_j), \qquad h_j \in \mathrm{span}\{B\text{-splines}\}, \tag{7}$$

by ridge regression on two thirds of a fresh 3,000-curve probe set, and define

$$\mathcal{A} = R^2(s, \hat{s}) \ \text{ on the held-out third.} \tag{8}$$

$\mathcal{A} = 1$ means the model is exactly additive over the probe distribution; $\mathcal{A} \approx 0$ means its behaviour is dominated by interactions no additive explanation can capture.

We are explicit about what (8) does and does not measure. It is reproducible, applied identically to every model, and is a genuine property of the learned function. It does *not* certify that an explanation is causally correct, stable under distribution shift, physically meaningful, or useful to a human. It measures one dimension of interpretability, and the sanity check that the linear model scores $\mathcal{A} = 0.999$ confirms the measurement behaves as intended.

4 TASK 1: DETECTION

Table 1 and Figure 2(a) give held-out ROC-AUC together with $\mathcal{A}$. Every family exceeds AUC = 0.90; the problem is not hard in aggregate, and the differences live at low SNR.

Two results are worth isolating. First, the LSTM is the weakest model (AUC = 0.906) despite being the family nominally designed for sequences. A transit is a localised, translation-tolerant shape; convolution and even a fixed linear template match that structure better than a recurrence that must carry information across 201 steps. Second, the KAN reaches AUC = 0.943, within 0.003 of the linear model, while retaining direct readability. Because Eq. (6) sums learned per-bin functions, the model's sensitivity profile across orbital phase can be plotted and compared against the true mean transit template; in the source study that comparison yields a correlation of 0.94, i.e. the KAN independently rediscovered the transit shape rather than latching onto an artefact.

Table 1. Detection on the held-out synthetic set. $\mathcal{A}$ is additive-surrogate fidelity, Eq. (8). “Readable” marks models whose learned function can be inspected directly without a surrogate.

Model	Family	ROC-AUC	$\mathcal{A}$	Readable
MLP	neural net	**0.975**	0.573	no
XGBoost	tree	0.964	0.822	no
Random forest	tree	0.959	0.903	no
CNN	neural net	0.956	0.768	no
Logistic/Ridge	linear	0.946	**0.999**	**yes**
KAN	KAN	0.943	0.900	**yes**
LSTM	neural net	0.906	0.728	no

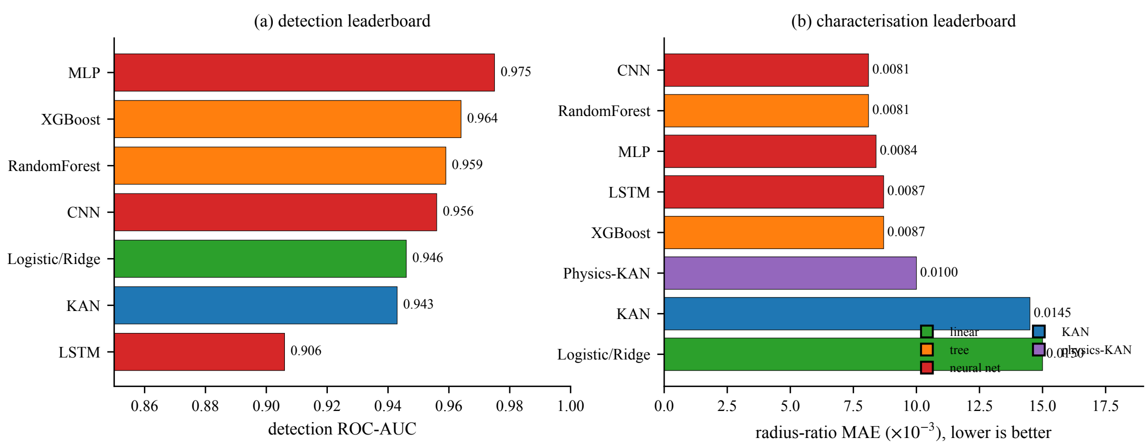

Fig. 2. (a) Detection ROC-AUC and (b) characterisation MAE in R_p/R_*, coloured by model family. Note the axis scales: the detection spread is 0.069 AUC across all seven families, whereas the characterisation spread is nearly a factor of two in error.

5 TASK 2: CHARACTERISATION

5.1 A label-free physics-informed estimator

Supervised regression of ρ requires labelled radii, which for real systems come from published fits and are therefore scarce and heterogeneous. We remove that dependence. A KAN maps the light curve to four bounded physical parameters via a sigmoid reparameterisation,

$$\theta_k = \ell_k + (u_k - \ell_k)\,\sigma(z_k(\mathbf{x}))\,, \qquad \theta = (\rho, b, a/R_*, t_0), \tag{9}$$

those parameters are pushed through the differentiable forward model (3), and the network is trained on the reconstruction objective alone:

$$\mathcal{L}_{\text{phys}} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{L}\sum_{j=1}^{L}\Big(F_{\theta(\mathbf{x}_i)}(\phi_j) - 1 - o_{ij}\Big)^2, \tag{10}$$

where o_{ij} is the median-subtracted observed flux. No radius label appears anywhere in (10). The network must infer ρ because ρ is the only way to reproduce the observed depth—the same inverse-problem logic that underlies physics-informed neural networks [11].

Table 2. Characterisation of R_p/R_* on the held-out synthetic set. Skill is $1 - \mathrm{MAE}/\Delta R_p$ with $\Delta R_p = 0.12$ the sampled range. The physics-KAN is the only entry trained without radius labels.

Model	Family	MAE	Skill	$\mathcal{A}$	Labels used
Random forest	tree	**0.0081**	0.933	0.476	yes
CNN	neural net	0.0081	0.933	0.426	yes
MLP	neural net	0.0084	0.930	0.298	yes
XGBoost	tree	0.0087	0.928	0.314	yes
LSTM	neural net	0.0087	0.928	0.484	yes
Physics-KAN	physics-KAN	0.0100	0.917	0.530	**no**
KAN	KAN	0.0145	0.879	0.989	yes
Logistic/Ridge	linear	0.0150	0.875	**0.997**	yes

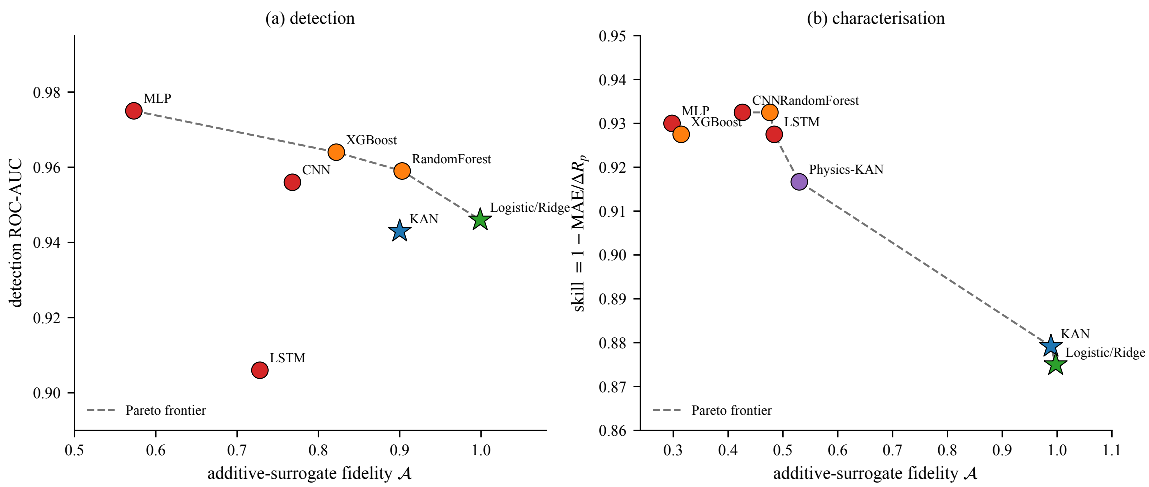

Fig. 3. Accuracy against additive-surrogate fidelity. Stars mark directly readable models, circles black boxes; the dashed line is the Pareto frontier. On detection (a) the frontier is short and the cost of transparency is 0.029 AUC. On characterisation (b) the frontier is long: full readability costs 40% higher error than the best black box, while the physics-informed model occupies an intermediate position.

5.2 Results

Table 2 and Figure 2(b) give the leaderboard. The best supervised model (random forest) reaches MAE = 0.0081. The physics-informed KAN reaches 0.0100—24% worse, while using *zero* radius labels—and beats both label-supervised readable models, the KAN (0.0145) and ridge (0.0150), by a wide margin.

The physics-informed model's advantage over the readable supervised models is structural. Ridge and the plain KAN must both express ρ as an additive function of per-bin flux, but ρ enters the forward model through Eq. (3) nonlinearly and jointly with b and a/R_*: depth alone does not determine radius when the geometry is unknown. The physics-KAN does not have to learn that coupling from data, because the coupling is built into its decoder.

5.3 The Pareto view

Figure 3 places every model on the accuracy–additivity plane. Three observations follow.

Table 3. Zero-shot recovery of R_p/R_* on twelve confirmed Kepler planets. "Screen" is the label-free reconstruction-residual rule: ✓ retained, × rejected as suspected model mismatch. The screen rejects Kepler-1624 b despite an error of 0.001.

Planet	Published	Recovered	$\lvert\Delta\rvert$	Screen
Kepler-719 b	0.080	0.081	0.001	✓
Kepler-1624 b	0.109	0.108	0.001	×
Kepler-855 b	0.074	0.081	0.007	✓
Kepler-550 b	0.049	0.060	0.011	✓
Kepler-546 b	0.056	0.069	0.013	✓
Kepler-228 d	0.040	0.055	0.015	✓
Kepler-734 b	0.042	0.060	0.018	✓
Kepler-170 b	0.030	0.052	0.022	✓
Kepler-244 b	0.031	0.054	0.023	✓
Kepler-485 b	0.121	0.076	0.045	×
Kepler-548 b	0.122	0.070	0.052	✓
Kepler-12 b	0.119	0.046	0.073	×
Full-sample MAE (primary)			**0.023**	
Screened-subset MAE (exploratory, $n = 9$)			0.018	

First, the trade-off is real but its exchange rate is task dependent. On detection, moving from the MLP ($\mathcal{A} = 0.573$) to the fully readable linear model ($\mathcal{A} = 0.999$) costs 0.029 AUC—cheap. On characterisation, the same move costs 85% higher error ($0.0081 \rightarrow 0.0150$)—expensive.

Second, high $\mathcal{A}$ does not imply high accuracy and low $\mathcal{A}$ does not imply the reverse. The MLP has both the best detection AUC and nearly the worst additivity, but the random forest achieves $\mathcal{A} = 0.903$ at AUC $= 0.959$—nearly linear-model transparency at nearly MLP accuracy. Assuming a strict frontier between the two axes is not warranted by these data.

Third, the physics-informed model is not on the Pareto frontier of Figure 3(b) and this is the correct result: on *these* axes it is dominated. Its distinguishing property is orthogonal to both—it requires no labels—which is precisely the situation where a two-axis Pareto plot is an incomplete decision aid.

6 ZERO-SHOT TRANSFER TO KEPLER

The physics-informed model was trained exclusively on simulation and has never seen a real light curve. We apply it, without adaptation, to twelve confirmed Kepler planets whose long-cadence photometry has been folded on the published orbital period and binned to the same 201-point grid [7].

6.1 Reading the result honestly

The full twelve-planet MAE is 0.023, and that is the primary number. It is roughly 2.3× the model's synthetic error, which is the expected penalty for sim-to-real transfer: real folds carry detrending residuals, contamination from neighbouring stars, and—most importantly—30-minute long-cadence integration, which smears short ingress and egress and systematically shallows the profile.

That smearing explains the dominant error pattern in Figure 4(b). The three largest published radii (Kepler-485 b, Kepler-548 b, Kepler-12 b, all $\rho > 0.12$) are underestimated by 0.045–0.073, whereas the small-radius planets are mildly

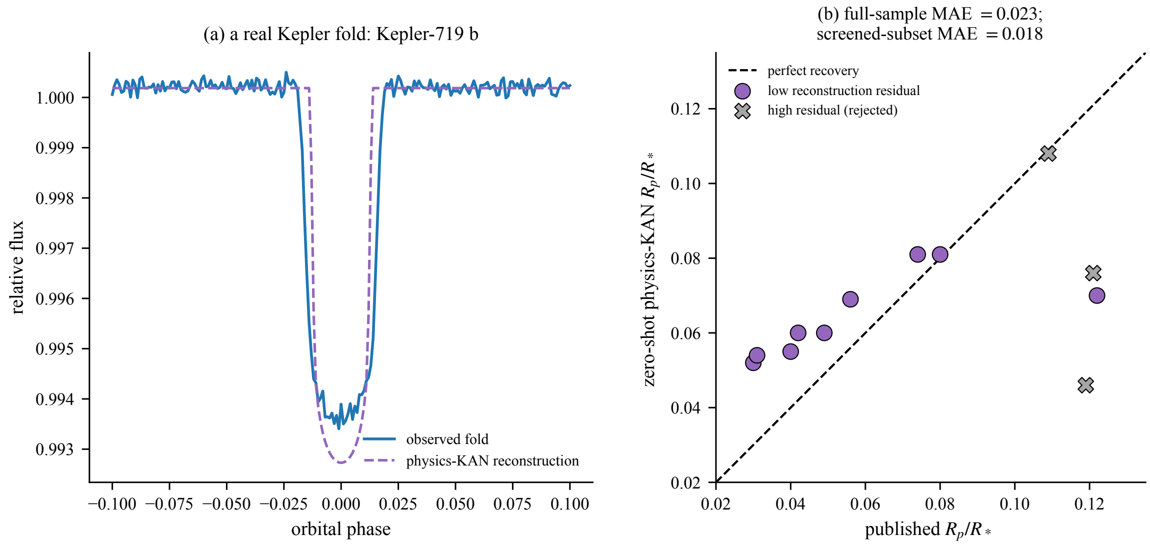

Fig. 4. (a) An observed Kepler fold with the physics-KAN's reconstruction overlaid. (b) Recovered against published radius ratio for all twelve planets; markers indicate the label-free reconstruction-residual screen. The three largest published radii are all underestimated, the signature of long-cadence smearing.

*over*estimated. Deep transits of short duration lose the most to integration, so this is a forward-model mismatch, not random error.

6.2 The residual screen does not track accuracy

The source study screens folds by reconstruction residual, a label-free diagnostic for model mismatch. Replicating that rule exactly, retaining the lowest-residual 75% improves MAE from 0.023 to 0.018—a modest gain, not the dramatic improvement a threshold chosen after seeing the data might suggest.

More informative is which folds the screen selects. It correctly rejects Kepler-12 b (error 0.073) and Kepler-485 b (error 0.045), but it also rejects Kepler-1624 b, whose radius was recovered to within 0.001—the joint-best result in the sample—while retaining Kepler-548 b, whose error of 0.052 is the second worst. Reconstruction residual measures how well the assumed forward model explains the fold, which is not the same quantity as how accurately ρ was recovered; a fold can be noisy yet yield a good depth estimate, or clean yet be fit with a compensating combination of b and a/R_*. Any confirmatory study should fix the quality threshold in advance and report accepted and rejected cases separately, as we have done here.

7 DISCUSSION

7.1 Supported findings

(1) Accuracy and additive-surrogate fidelity are distinct objectives with a task-dependent exchange rate: 0.029 AUC for detection, 85% relative error for characterisation.

(2) A differentiable transit model permits training a radius estimator by reconstruction alone, within 24% of fully supervised accuracy and far ahead of supervised readable models.

(3) Zero-shot transfer to real Kepler folds succeeds qualitatively (MAE = 0.023) and fails in a physically interpretable direction, namely underestimation of deep transits under long-cadence smearing.
(4) Reconstruction residual is a poor proxy for parameter accuracy and should not be used as a post-hoc accuracy filter.

7.2 Threats to validity

Training and test simulations share a generator, parameter ranges, and noise model, so synthetic scores measure interpolation within a known family rather than robustness to unmodelled systematics. The physics-informed model enjoys a matched-model advantage: its decoder is the very process that generated the synthetic data, and the Kepler results show how much of that advantage survives when the match is broken. Additive-surrogate fidelity captures one dimension of interpretability and is measured on a probe distribution drawn from the same simulator. Twelve planets is a demonstration, not a population estimate, and the sample is biased towards well-characterised systems. Finally, no result is repeated across training seeds, so the reported differences carry unquantified optimisation variance; differences smaller than roughly 0.005 AUC or 0.0005 in MAE should not be treated as meaningful.

7.3 Future work

The most informative next experiment is a mismatch study: train on the current simulator and test under correlated noise, unmodelled limb darkening, and explicit long-cadence integration, which Figure 4 identifies as the dominant real-data error source. Adding the integration kernel to Eq. (3) is straightforward and should directly address the deep-transit bias. Extending to TESS data [13] and to the single-transit regime, where folding is unavailable and physics-informed inference should have its largest relative advantage [3], would test the approach where it matters most.

8 CONCLUSION

On simulated transits, the most accurate detector is opaque and the most transparent one gives up 0.029 AUC; on radius measurement the same transparency costs 85% more error. There is no single best model, and the choice depends on whether raw performance or a checkable explanation matters more for the scientific question at hand. Between those poles sits a third option that the two-axis framing does not capture: a physics-informed estimator that reaches near-supervised accuracy using no parameter labels at all, and whose errors on real data are interpretable as a specific, correctable mismatch between the assumed forward model and the instrument.

REPRODUCIBILITY

The figure script accompanying this manuscript regenerates every figure and table, including the forward model of Eq. (3) and the label-free residual screen, from the transcribed leaderboards and the embedded twelve-planet Kepler sample. The source notebook validates the differentiable forward model against `batman` [6] and fixes all data-generation seeds.

REFERENCES

[1] William J. Borucki, David Koch, Gibor Basri, Natalie Batalha, Timothy Brown, Douglas Caldwell, et al. 2010. Kepler Planet-Detection Mission: Introduction and First Results. *Science* 327, 5968 (2010), 977–980.
[2] Antonio Claret. 2000. A New Non-Linear Limb-Darkening Law for LTE Stellar Atmosphere Models. *Astronomy and Astrophysics* 363 (2000), 1081–1190.

[3] Daniel Foreman-Mackey, Timothy D. Morton, David W. Hogg, Eric Agol, and Bernhard Schölkopf. 2016. The Population of Long-Period Transiting Exoplanets. *The Astronomical Journal* 152, 6 (2016), 206.

[4] Trevor J. Hastie and Robert J. Tibshirani. 1990. *Generalized Additive Models*. Chapman and Hall.

[5] David M. Kipping. 2013. Efficient, Uninformative Sampling of Limb Darkening Coefficients for Two-Parameter Laws. *Monthly Notices of the Royal Astronomical Society* 435, 3 (2013), 2152–2160.

[6] Laura Kreidberg. 2015. batman: BAsic Transit Model cAlculatioN in Python. *Publications of the Astronomical Society of the Pacific* 127, 957 (2015), 1161–1165.

[7] Lightkurve Collaboration. 2018. Lightkurve: Kepler and TESS Time Series Analysis in Python. *Astrophysics Source Code Library, ascl:1812.013* (2018).

[8] Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, and Max Tegmark. 2024. KAN: Kolmogorov–Arnold Networks. *arXiv preprint arXiv:2404.19756* (2024).

[9] Kaisey Mandel and Eric Agol. 2002. Analytic Light Curves for Planetary Transit Searches. *The Astrophysical Journal Letters* 580, 2 (2002), L171–L175.

[10] Kyle A. Pearson, Leon Palafox, and Caitlin A. Griffith. 2018. Searching for Exoplanets Using Artificial Intelligence. *Monthly Notices of the Royal Astronomical Society* 474, 1 (2018), 478–491.

[11] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. 2019. Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations. *J. Comput. Phys.* 378 (2019), 686–707.

[12] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 1135–1144.

[13] George R. Ricker, Joshua N. Winn, Roland Vanderspek, David W. Latham, Gáspár Á. Bakos, Jacob L. Bean, et al. 2015. Transiting Exoplanet Survey Satellite (TESS). *Journal of Astronomical Telescopes, Instruments, and Systems* 1, 1 (2015), 014003.

[14] Cynthia Rudin. 2019. Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead. *Nature Machine Intelligence* 1, 5 (2019), 206–215.

[15] Christopher J. Shallue and Andrew Vanderburg. 2018. Identifying Exoplanets with Deep Learning: A Five-Planet Resonant Chain Around Kepler-80 and an Eighth Planet Around Kepler-90. *The Astronomical Journal* 155, 2 (2018), 94.

AUTHORS' BACKGROUND

Your Name	Title*	Research Field	Personal website
Bhavya Gondra		Astronomical data analysis	
Akhil Muvvala		Astronomical data analysis	
Milan Petrov		Astronomical data analysis	
Harish Senthil		Astronomical data analysis	
Ayan Sharma		Astronomical data analysis	
Manav Sharma		Astronomical data analysis	
Pranavan Sivaraman		Astronomical data analysis	
Adhrit Somayajula		Astronomical data analysis	
Srikanth Samy		Astronomical data analysis	
Ishaan Gupta		Astronomical data analysis	

*This form helps us to understand your paper better, the form itself will not be published.

*Title can be chosen from: master student, Phd candidate, assistant professor, lecture, senior lecture, associate professor, full professor